DATA STRUCTURES — II

How we look things up and represent connections.

SCAN & LEARN

WHAT HAPPENS • FOLLOW THE ARROWS

01 WHAT IS THE TASK? → 02 PICK A STRUCTURE → 03 PERFORM AN ACTION → 04 CONSIDER THE COST

FOUR IDEAS TO KEEP

1 DICTIONARY = KEY → VALUE

Look up an item using a KEY, not a position. Examples: name → phone; product → price. Called dict in Python.

{"Ana": 12, "Dan": 9} → key “Ana” gives 12.

2 TREE = FAMILY OF BRANCHES

Each node (except the root) has one parent. Computer folders are often organized like a tree.

University → Year 1 → Semester 1 → Lab.

3 GRAPH = DOTS AND LINKS

Nodes connected by edges. Friend networks and metro routes are graphs. Graphs can have cycles.

A — B — C and A — C: more than one path exists.

4 HOW DO YOU CHOOSE?

Order → list. Oldest first → queue. Newest first → stack. Lookup by key → dict. Relations → graph.

The right structure simplifies a program and can save work.

TRY IT — DRAW CONNECTIONS

On paper: draw 5 towns and roads between them. Can you reach any town from any other?

THE ONE SENTENCE TO REMEMBER

A DATA STRUCTURE is the RULE for storing and finding information.

Question: What structure fits “student name → grade”?